

Stable Distributions

What limits can appear for normalized sums of i.i.d. random variables?

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Definitions of stable distributions

The following definitions of stable distributions are equivalent

Definition

A non-degenerate random variable X is called stable if, for any $a, b > 0$, there exist constants $c > 0$ and $d \in \mathbb{R}$ such that

$$aX_1 + bX_2 \stackrel{d}{=} cX + d, \quad (1)$$

where X_1 and X_2 are independent copies of X .

Definition

A non-degenerate random variable X is stable if and only for every $n \in \mathbb{N}, n \geq 2$, there exist constants $c_n > 0$ and $d_n \in \mathbb{R}$ such that

$$X_1 + \cdots + X_n \stackrel{d}{=} c_n X + d_n \quad (2)$$

where $X_1, \dots, X_n$ are independent copies of X .

Definition

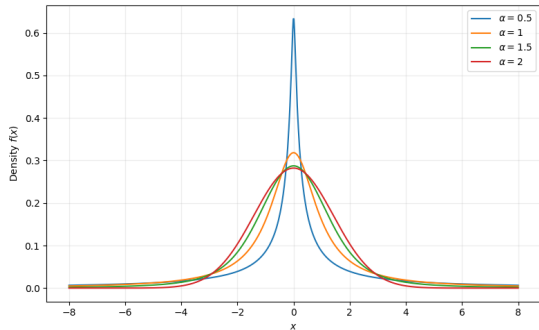
A non-degenerate random variable X is stable if and only if $X \stackrel{d}{=} \gamma Z + \delta$, where $0 < \alpha \leq 2$, $-1 \leq \beta \leq 1$, $\gamma > 0$, $\delta \in \mathbb{R}$ and Z is a random variable with characteristic function

$$\mathbb{E} \exp(iuZ) = \begin{cases} \exp(-|u|^\alpha [1 - i\beta \tan(\frac{\pi\alpha}{2}) \operatorname{sign}(u)]), & \alpha \neq 1, \\ \exp(-|u| [1 + i\beta \frac{2}{\pi} \operatorname{sign}(u) \log |u|]), & \alpha = 1. \end{cases} \quad (3)$$

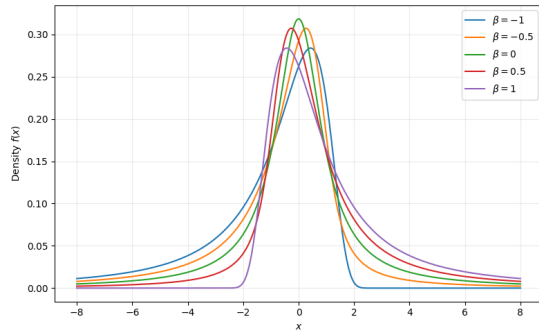
We parametrize stable distributions through the characteristic function of the standard stable variable Z .

- α - stability index
- β - skewness parameter
- γ - scale parameter
- δ - location parameter

Effect of parameters



Fixed $\beta = 0$, varying α .



Fixed $\alpha = 1$, varying β .

Generalized Central Limit Theorem

Theorem (Generalized Central Limit Theorem)

A non-degenerate random variable Z is stable for some $0 < \alpha \leq 2$ if and only if there is an i.i.d. sequence of random variables X_n and constants $a_n > 0, b_n \in \mathbb{R}$ with

$$a_n(X_1 + \cdots + X_n) - b_n \xrightarrow{d} Z$$

Proof of GCLT

We need convergence of types theorem:

Theorem (Convergence of Types theorem)

Let X_n be a sequence of random variables and $(a_n)_{n \geq 1} > 0$, $(b_n)_{n \geq 1}$ be a sequence of real constants. Assume

$$X_n \xrightarrow{d} X \quad \text{and} \quad a_n X_n + b_n \xrightarrow{d} Y$$

where neither X nor Y is degenerate. Then $a_n \rightarrow a > 0$ and $b_n \rightarrow b$ and $aX + b \stackrel{d}{=} Y$

Board proof:

limit of normalized sums $\implies$ stability of the limit.

The converse follows directly from the definition of stability.

Proof of GCLT Continued

Set

$$A_n = \frac{a_n}{a_{kn}}, \quad B_n = \frac{a_n b_{kn}}{a_{kn}} - kb_n.$$

Then

$$\begin{aligned} A_n Y_n + B_n &= \frac{a_n}{a_{kn}} (a_{kn} S_{kn} - b_{kn}) + \left(\frac{a_n b_{kn}}{a_{kn}} - kb_n \right) \\ &= a_n S_{kn} - kb_n. \end{aligned}$$

We get

$$\begin{aligned} A_n Y_n + B_n &= a_n (T_1(n) + \cdots + T_k(n)) - kb_n \\ &= (a_n T_1(n) - b_n) + \cdots + (a_n T_k(n) - b_n). \end{aligned}$$

Definition

A random variable X is in the domain of attraction of Z if there exist constants $a_n > 0, b_n \in \mathbb{R}$ with

$$a_n(X_1 + \cdots + X_n) - b_n \xrightarrow{d} Z$$

where $X_1, X_2, X_3, \dots$ are independent and identically distributed copies of X . $DA(Z)$ is the set of all random variables that are in the domain of attraction of Z .

Gaussian domain of attraction

Theorem (Gaussian domain of attraction criterion)

A random variable X is in the domain of attraction of a Gaussian law if and only if

$$\frac{t^2 \int_{|x|>t} dP_X(x)}{\int_{|x|<t} x^2 dP_X(x)} \rightarrow 0, \quad t \rightarrow \infty.$$

Board proof: finite, positive variance implies the criterion, and hence recovers the classical CLT case.

Domain of attraction: tail criterion

Definition (Slowly varying function)

A function L is slowly varying at infinity if, for every fixed $a > 0$,

$$\frac{L(ax)}{L(x)} \rightarrow 1 \quad \text{as } x \rightarrow \infty.$$

Theorem (Tail criterion)

Let $0 < \alpha < 2$. Then

$$X \in DA(Z(\alpha, \beta))$$

if and only if, for some $c^+, c^- \geq 0$, $c^+ + c^- > 0$, and slowly varying L ,

$$\mathbb{P}(X > x) \sim c^+ x^{-\alpha} L(x), \quad \mathbb{P}(X < -x) \sim c^- x^{-\alpha} L(x).$$

In this case,

$$\beta = \frac{c^+ - c^-}{c^+ + c^-}.$$

Example: Pareto distribution

Let

$$X \sim \text{Pareto}(\theta, 1), \quad F(x) = 1 - x^{-\theta}, \quad x \geq 1.$$

Three cases:

$$X \in \begin{cases} DA(Z(\theta, 1)), & 0 < \theta < 2, \\ DA(Z(2)), & \theta = 2, \\ DA(Z(2)), & \theta > 2. \end{cases}$$

- $0 < \theta < 2$: non-Gaussian stable limit.
- $\theta = 2$: Gaussian borderline case with infinite variance.
- $\theta > 2$: finite variance, hence classical CLT.

The rest of the project studies further structural properties, especially tail behaviour, unimodality and infinite divisibility.